

# Designing Effective Output

CSE 4407

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# Output: The 'Face' of Your System

- What
  - Information delivered to users by the system
  - **Channels:** Intranets, Extranets, Web, Mobile Apps, etc.
- Forms
  - **Soft Copy:** Screen displays (desktop, tablet, smartphone), video, audio, social media
  - **Hard Copy:** Printed reports
- **Key Idea:** Users often judge the *entire system's* merit based on its output

# The Goal: Creating Useful Output

- **Why:** Useful output drives system adoption and user satisfaction
- **Analyst's Role:** Collaborate *closely* with users
- **Focus:** Six fundamental objectives guide the design of effective output
  - Serve the intended purpose
  - Fit the user
  - Deliver the appropriate quantity
  - Ensure output is where it is needed
  - Provide output on time
  - Choose the right output method

# Designing Output: Purpose, Fit, and Quantity

- **Serve the intended purpose**
  - Align output directly with user/organization goals
  - If output is not functional or purposeful, do not create it
- **Fit the user**
  - Tailor output to the user's tasks, role, and preferences
  - Personalization/customization is key, especially for interactive systems
  - Consider user needs identified via interviews, observation, prototypes
- **Deliver appropriate quantity**
  - Provide what's needed to complete work, *but...*
  - Avoid "Information Overload"
  - Consider showing subsets initially, with options to drill down or link for more details.

# Delivering Output: Time, Location, and Method

- Ensure output is where it is needed
  - Distribution is crucial, even with digital output
  - Output has no value if it doesn't reach the right decision-maker
- Provide output on time
  - Timeliness is critical for decision-making → Late information = Useless information
  - Consider required frequency (Daily, weekly, monthly, real-time, by exception?)
  - Web-based output can improve timely access
- Choose the right output method
  - Screen, print, audio, video, file export? → Consider cost, accessibility, flexibility, durability, distribution, and impact trade-offs
  - Choice requires careful consideration

# Content Meets Method

- **Key:** The *what* (content) influences the *how* (method)
  - Function dictates form
  - Intended purpose guides the choice of output delivery
- **Think Broadly:** Any useful information generated by the system is output
- **Categories**
  - **External Output:** Goes *outside* the business
    - **Example:** Public website content, customer invoices, shareholder reports
    - **Characteristics:** May need instructions, often uses official branding (logo, color), placed on preprinted forms or designed web templates
    - **Trend:** Encouraging electronic versions (online bills) for Green IT and efficiency
  - **Internal Output:** Stays *within* the business
    - **Example:** Intranet pages, summary reports for managers, detailed operational logs, exception reports
    - **Types:** Summaries, detailed, historical, exception-based

# Output Technologies Overview

| Modality       | Example                                   | Advantages                                                                  | Disadvantages                                                        |
|----------------|-------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| Printer        | Standard, specialized, 3D                 | Affordable, flexible, high volume, reliable                                 | Needs operator, compatibility issues, supply costs, slow, less green |
| Display        | Monitor, integrated displays, touchscreen | Interactive, real-time, quiet, good for frequent/brief info, uses PC power  | Needs space/cabling, save screen views                               |
| Audio          | Loudspeakers, headphones, earbuds         | Good for individuals, transient messages, hands-free use, wide distribution | Needs earbuds for privacy/focus, limited application                 |
| Mobile Devices | Smartphones                               | Portable, interactive, zoomable                                             | Small screen, confusing icons/buttons, easily lost/stolen            |
| Electronic     | Web pages, email, RSS feeds, blogs, APIs  | Reduces paper, easily updated, broadcast possible, interactive              | Formatting limits, context loss, maintenance                         |

# Choosing the Right Technology: Key Factors

- **Balancing Act:** Selecting the best tech involves considering many interrelated factors
- **Key Questions**
  - Who will use it?
  - How many people need it?
  - Where is it needed?
  - What is the purpose?
  - What speed is required?
  - How frequently accessed?
  - How long to store it?
  - What regulations apply?
  - What are the costs?
  - What environmental needs?

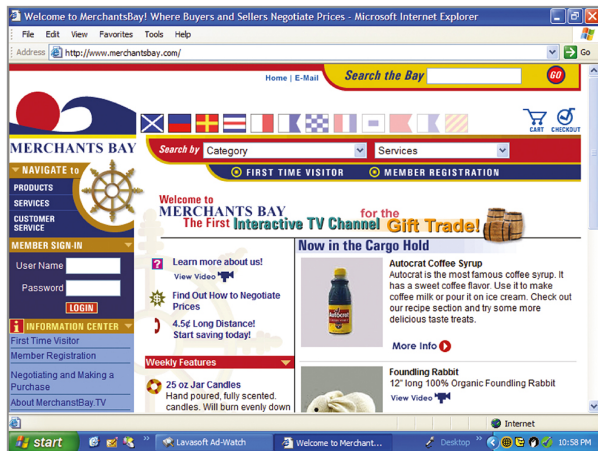
# Understanding Output Users and Quality Needs

- **Who Will Use the Output?**

- **User Profile**

- Job role, technical skill, location, mobility
    - E.g., mobile manager needs portable output vs. deskbound dispatcher needs screen output

- **Internal vs. External:**  
Different needs, different presentation



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# Output Volume and Distribution Considerations

- **How Many People Need It?**

- Large audience

- Web documents (easily accessible, scalable) with print options
    - Consider broadcast methods like email lists

- Small audience: Screen display, maybe even audio

- **Where Is Output Needed?**

- Local use? → Intranet pages or local printing
  - Distributed/Remote users?
    - Web or Extranets
    - Electronic distribution
  - Regulations? → Physical copy

# Purpose and Timeliness of Output

- **What is the Purpose?**

- Attract investors? → Polished Annual Report (print and web)
- Document history? → Video/Multimedia
- Provide instant update? → Screen crawls, dynamic web content, audio alerts
- Enable user analysis? → Output with embedded tools
- Support Org Processes? → Tracking dashboards, schedules, monitoring alerts

- **What Speed is Needed?**

- **Operational Level**
  - Needs rapid output for immediate action
  - Online screens are vital
- **Strategic Level**
  - Needs periodic output for analysis, forecasting, trends
  - Timed reports

# Access, Storage, and Regulatory Requirements

- **How Frequently Accessed?**

- High frequency
  - Web/Online systems are ideal
  - Easy access, avoids wear-and-tear on paper
- Low frequency: Archiving (digital, cloud-based)

- **How Long to Store It?**

- Paper degrades → Digital lasts longer, but requires accessible technology
- Driven by government regulations, internal policies, historical value
- Cloud storage offers solutions for long-term digital retention

- **What Special Regulations Apply?**

- Government rules (local, state, federal) influence
  - Output content
  - Storage requirements
  - Distribution rules
- Industry-specific rules

# Cost and Environmental Considerations

- **What are the Costs?**

- **Initial Costs:** Hardware purchase/lease (printers, screens), software licenses, development time, web hosting setup
- **Ongoing Costs:** Supplies (paper, ink/toner), maintenance contracts, electricity, website updates, hosting fees, staff time
- **Key:** Look beyond the purchase price!

- **Human and Environmental Requirements?**

- **Human:** Accessibility, ease of use, training needs
- **Physical Environment:** Space for equipment, cabling, Wi-Fi coverage, temperature/humidity control, noise considerations
- **Green IT/Sustainability:** Minimize paper, encourage digital workflows, energy efficiency of devices

# Output Isn't Neutral: Understanding Bias

- **Key Concept:** Output actively shapes user perception and decision-making
- **Analyst's Impact:** Choice of data, ordering, wording steers interpretation
- **Bias is Everywhere:** Every report or chart reflects some perspective
- **Goal:** Not to eliminate bias entirely (impossible!), but to:
  - Avoid presenting information in a misleading way
  - Inform users about any remaining limitations or biases

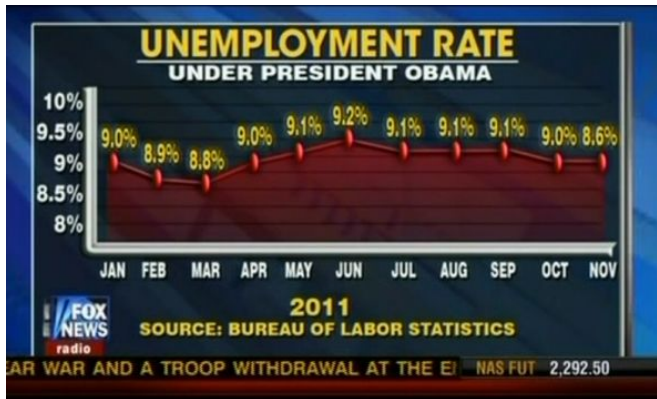
# Sources of Unintentional Output Bias

- Common Culprits

- **Sorting:** The order in which information is presented
  - **Primacy Effect:** First items grab attention
  - **Alphabetic Sort:** Emphasizes items starting with A, B, ... → 'Aardvark Printers' get undue attention over 'Zephyr Supplies'
  - **Other Sort:** Chronological or Cost-based sorting can also introduce bias depending on context → SABRE and Apollo reservations systems in 1980s
- **Setting Limits:** Defining thresholds, especially for exception reports
  - **Limits Too Low:** Generates too many "exception," making problems seem bigger or more frequent than they are
  - **Limits Too High:** Real issues might be missed
- **Choice of Graphics:** How data is visualized (type, scale, color, size)

## Bias through Graphics

- Visualizations are powerful ...and powerfully misleading if not done carefully
- Bias can come from graph type, size, color choice, and especially scale



# Designing for Fairness: Minimizing Output Bias

- **Self-Check Assumption:** Use a bias-review checklist before finalizing design
- **Involve Real Users**
  - Prototype interactions; gather feedback
  - Test across devices, browsers, screen sizes
- **Educate and Warn**
  - Include brief notes on possible limits or biases
  - Offer “info” pop-ups explaining data scope
- **Build Flexibility:** Let users adjust sort order, filters, thresholds
- **Promote “Reality Tests”:** Encourage comparison of multiple reports or data sources

# Designing Printed Output: Types and Sources

- **Foundation:** The Data Dictionary → Defines data element names, required field lengths, formats, etc.
- **Types of Printed Reports**
  - **Detailed Reports:** Prints info for *every* record on the relevant file
    - **Use Cases:** Customer mailings, product catalogs, student grade lists
    - **Note:** Many detailed reporting needs are now met by on-screen inquiries
  - **Exception Reports:** Prints info *only* for records meeting specific criteria (the “exceptions”)
    - **Use Cases:** Identifying items needing reorder, students on the Dean’s list, accounts significantly overdue
    - **Critical Link:** Requires careful thought about setting limits
    - Primarily used for operational control
  - **Summary Reports:** Prints one line representing a *group* of records (e.g., totals, averages)
    - **Use Cases:** Analyzing sales trends, department performance overviews
    - Primarily used for management decision-making

# Don't Forget Print!

- **Reality Check:** Users will print web pages
- **Design for Printability**
  - Offer downloadable PDFs of key documents
  - Test direct printing in Chrome, Firefox, Edge, Safari, etc.
  - Apply `@media print` CSS to hide menus, tweak fonts/colors
- **Goal:** Clean, readable, professional output

# Designing Output for Displays

- Difference from Print
  - Not permanent like paper
  - Can be personalized more easily
  - Available on demand
  - Tied to a device
  - Interactive
- Key Considerations
  - Clear navigation instructions
  - Access control

# Guidelines for Display Design

- Core Guidelines

- Keep it Simple

- Avoid clutter
    - Show only what's needed
    - Make info easy to scan

- Stay Consistent: Use uniform layouts, terms, and navigation

- Facilitate Movement

- Enable easy navigation
    - Allow drill-down and backtracking

- Make It Visually Appealing

- Use whitespace, alignment, and clear fonts
    - Pick readable colors

- Get User Feedback

- Use prototypes
  - Iterate based on real input

# Example: Order Status Display

- Simple Layout
  - Shows key status info only
  - No visual clutter
- Consistent (Implied): Sample style used across status screens
- Supports Navigation: Clear instructions
- Visually Clear
  - Bold heading
  - Tabular view for easy reading
  - Includes context for user

| New Zoo Order Status |         |            |                     |
|----------------------|---------|------------|---------------------|
| Retailer             | Order # | Order Date | Order Status        |
| Animals Unlimited    | 933401  | 09/05/2022 | Shipped On 09/29    |
|                      | 934567  | 09/11/2022 | Shipped On 09/21    |
|                      | 934613  | 09/13/2022 | Shipped On 09/21    |
|                      | 934691  | 09/14/2022 | Shipped On 09/21    |
| Bear Bizarre         | 933603  | 09/02/2022 | Partially Shipped   |
|                      | 933668  | 09/08/2022 | Scheduled For 10/03 |
|                      | 934552  | 09/18/2022 | Scheduled For 10/03 |
|                      | 934683  | 09/18/2022 | Shipped On 09/28    |
| Cuddles Co.          | 933414  | 09/12/2022 | Shipped On 09/18    |
|                      | 933422  | 09/14/2022 | Shipped On 09/21    |
|                      | 934339  | 09/16/2022 | Shipped On 09/26    |
|                      | 934387  | 09/18/2022 | Shipped On 09/21    |
| Stuffed Stuff        | 934476  | 09/25/2022 | Backordered         |
|                      | 934341  | 09/14/2022 | Shipped On 09/26    |
|                      | 934591  | 09/18/2022 | Partially Shipped   |
|                      | 934633  | 09/26/2022 | Backordered         |
|                      | 934664  | 09/29/2022 | Partially Shipped   |

Press any key to see the rest of the list; ESC to end; ? for help.

For more detail place cursor over the order number and hit the Enter key.

# Example: Facilitating Movement

- **Facilitates Movement:** Users can drill down from summary to detail
- **Keeps Initial Screen Simple**
  - Only essential info shown at first
  - Prevents screen clutter
- **Consistent Instructions**
  - Same help/navigation cues at bottom
  - Builds user familiarity
- **Use Control**
  - Users decide when and if they want more detail
  - No forced navigation

| Order #                      | Retailer                                                |                 | Order Date    | Order Status      |                       |
|------------------------------|---------------------------------------------------------|-----------------|---------------|-------------------|-----------------------|
| 933603                       | Bear Bizarre<br>1001 Karhu Lane<br>Bern, Virginia 22024 |                 | 09/02/2022    | Partially Shipped |                       |
| Units                        | Pkg                                                     | Description     | Price         | Amount            | Detailed Status       |
| 12                           | Each                                                    | Floppy Bears    | 20.00         | 240.00            | Backordered Due 10/15 |
| 6                            | Each                                                    | Growlers        | 25.00         | 150.00            | Backordered Due 10/15 |
| 2                            | Each                                                    | Special Edition | 70.00         | 140.00            | Shipped 09/02         |
| 1                            | Box                                                     | Celebrity Mix   | 150.00        | 150.00            | Shipped 09/02         |
| 12                           | Each                                                    | Polar Bears     | 10.00         | 120.00            | Backordered Due 10/30 |
|                              |                                                         |                 |               | 800.00            |                       |
| Contact                      | Account Balance                                         |                 | Credit Rating | Last Order        | Shipped               |
| Ursula Major<br>703-484-2327 | 0.00                                                    |                 | Excellent     | 08/21/2022        | On Time               |

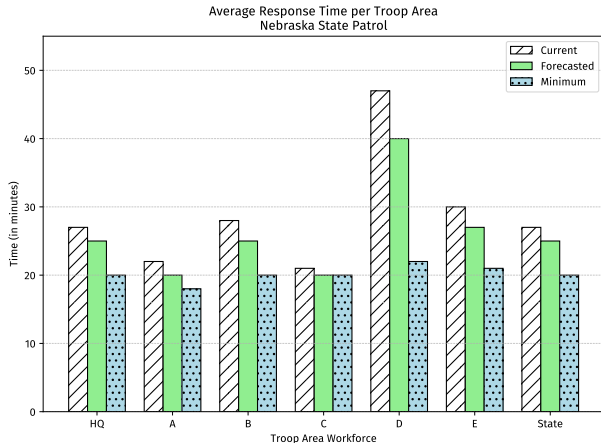
Press any key to see the rest of the list; ESC to end; ? for help.

# Using Graphical Output on Displays

- Why
  - Spot trends and patterns faster than with tables
  - Visually compare values at a glance
- Key Considerations
  - **Work with users:** Choose the *right* graph for message
  - **Keep it clear:** Must be easy to read and accurate
  - **Expose assumptions:** Explain scales or axes to avoid misinterpretation
- Design Factors
  - **Purpose:** Inform, compare, show trends
  - **Data Type:** Time series, categories, distributions
  - **Audience:** Experts vs. novices
  - **Perception:** Graph types influence how users interpret data

## Example: Nebraska Patrol DSS Graph

- **Intelligence:** Flags problem areas → Troops D and E have high response times
- **Design:** Compare current, forecast, and minimum times → See potential impact
- **Choice:** Support smart resource allocation



# Dashboards: Information at a Glance

- What

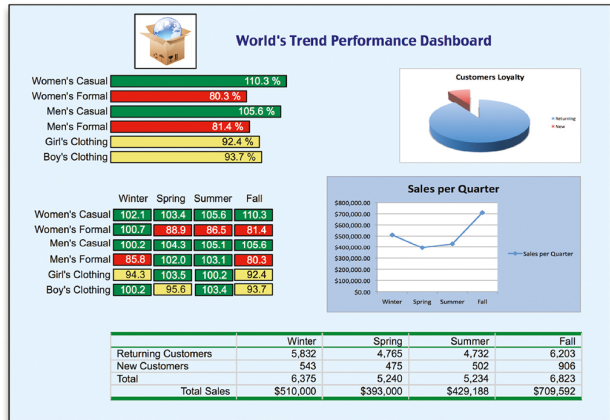
- Like a car dashboard → Multiple indicators, one screen
- Used by execs to monitor key metrics (KPIs) quickly
- Shows
  - Graphs and speedometer-like visuals
  - Warnings/alerts like engine lights
  - Key text/figures like odometers

- Goal

- Give a clear snapshot of performance
- Support fast decisions with digestible data
- But... only if well-designed

# Example: World's Trend Performance Dashboard

- All key views on one screen
- Helps manager assess performance from many angles
- Tools like Tableau make this possible



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# Dashboard Design Rules

- **Provide Context**
  - Don't just show a number
  - Show it relative to goals, history, or benchmarks
  - **Example:** Is \$851k sales good or bad?
- **Summarize Wisely**
  - Avoid excessive detail
  - Use summaries where appropriate
  - **Example:** \$851,235.32 vs. \$851k
- **Choose the Right Metrics:** *Variances* vs. Raw numbers
- **Stay Fair**
  - Avoid distortions
  - **Example:** Misleading scales
- **Correct Graph Style:** Line for trends, bar for comparison

## Dashboard Design Rules (Cont.)

- **Good Layout:** Clear labels, smart color user
- **Limit Chart Types:** Too much variety = Visual noise
- **Highlight Wisely:** Bold/red for key items
- **Group Logically:** Related metrics = Side by side
- **Keep it Clean:** No flashy logos or background art
- **One Screen:** Avoid scrolling or tabs
- **Be Flexible:** Gather feedback, iterate often

# Infographics: Visual Story telling

- What
  - A data story told visually
  - Combines numbers, charts, icons, and text
- When to Use
  - Engages audiences more than plain tables
  - Makes data fast and easy to grasp
  - Ideal for reports, marketing, or internal updates
- Tools: Web apps like Canva, Piktochart, etc.
- Example
  - Mixes standard and custom visuals
  - Clear message
  - Clean design with a limited palette



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# Infographics Design Tips

- **Start Strong:** Use a newsworthy headline
- **Tell a Story:** Arrange visuals with purpose
- **Mix Elements:** Charts, icons, maybe cartoons? → Skip busy photos
- **Keep It Brief:** Visuals carry the message
- **Limit Colors:** Stick to 2-4 coordinated shades
- **Call to Action:** Add contact info if relevant
- **Design to Share:** Perfect for social media

# Designing a Website: Beyond a Single Page

- From single “pages” to full “websites”
- First impression = landing page
- Web → An uncontrolled environment: Different
  - Browsers
  - Screen sizes/resolutions
  - Devices (PCs, tablets, phones)
  - User settings (fonts, JS)
- **Big Challenge:** Design for **variability**
- Builds on display design, but adds web-specific twists

# Designing for All Screens: Responsive and Adaptive

- Responsive Web Design (RWD)
  - One site, works on all devices
  - Uses fluid grids (percent-based layout)
  - Content reflows with screen
  - Ads may not be flexible → Use Ajax or alternate landing pages
- Adaptive Web Design (AWD)
  - Detects device, loads best layout
  - Uses fixed grids for desktop, tablet, and mobile
  - Offers more control across device types

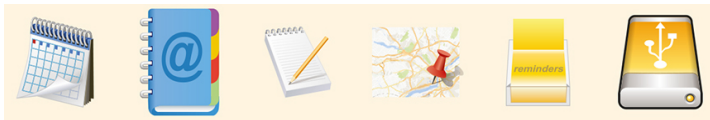
# Building Websites: Tools and Editors

- **Cloud-Based WYSIWYG:** Weebly, Wix
  - Drag and drop, no coding
  - Fast, easy, freemium model
  - **Downside:** Host lock-in, limited free features
- **Stand-Alone Software:** Dreamweaver
  - Full features, no ads
  - Own your files → Host anywhere
  - Costs more, steeper learning curve

# Design Trends: Skeuomorphic vs. Flat

- **Skeuomorphic Design**

- Mimics real-world objects
- Uses shadows, textures, fake realism
- Became outdated



- **Flat Design**

- Clean, minimal, 2D
- No shadows, simple shapes
- Focuses on clarity and function



- **Trends Evolve:** Flats may fade; 3D realism may return

# Preparing for Web Design: Tools, Research, and Inspiration

- **Use Design Tools**
  - Stand-alone (Dreamweaver) or cloud (Wix)
  - Faster and richer than hand-coding
- **Study Great Sites**
  - What works? What's engaging?
  - Structure, features, flow
- **Learn from Pros:** Critique design, interactivity, colors, metaphors
- **Evaluate with Tools:** Use **forms/checklists** to rate sites

# Bringing Ideas to Life: Planning, Prototyping, and Best Practices

- **Plan the Site First**
  - **Storyboards:** Screen flow like movie scenes
  - **Wireframes:** Basic layout → Boxes, no style
  - **Mockups:** Styled, realistic screens before coding
- **Use Templates:** Set layouts → Faster + Consistent design
- **Use CSS**
  - Separates content from style
  - One style change updates many pages
- **Be Cautious with Plug-Ins**
  - Flash, extensions → Not always supported
  - Stick to broadly compatible tech

# Specific Website Design Guidelines

- **Structure:** Logical organization and planning.
- **Content:** Valuable and engaging information/features.
- **Text:** Clear, readable, well-formatted writing.
- **Graphics:** Appropriate use of images.
- **Presentation Style:** Overall look, feel, and layout.
- **Navigation:** How users move through the site.
- **Promotion:** Making the site discoverable.

# Design Details: Structure and Content

- Structure
  - Define goals and page purposes
  - Plan site with mapping tools
  - Avoid broken links
- Content
  - **Provide Value:** Tips, tools, interactivity
  - Boost “stickiness” (time-on-site)
  - Use helpful metaphors (sparingly!)
  - Add FAQ page
  - Include widgets
- Examples: Polygon, Arngren Electronics

# Design Details: Text and Visuals

- Text

- Clear page titles
- Strong opening sentence
- Simple, scannable writing
- Keep content fresh

- Graphics

- Use PNG/JPEG/GIF wisely
- Simple backgrounds
- Small file sizes = Faster load
- Reuse elements for caching
- Add tooltips for images
- Test on multiple devices/screens

- Examples: [McMaster-Carr](#), [Berkshire Hathaway](#), [Yale School of Art](#)

## Design Details: Consistent and Fast Experience

- Fast home page (<3s, <100KB)
- Simple menus, clear options
- Limit heavy graphics
- Big, colorful headings
- Use image maps for creative links
- CSS for control and consistency
- Use <div>s, not tables, for layout
- Use JavaScript for interactivity, not annoyance
- Examples: [FormaFantasma](#), [James Bond Museum](#)

# Design Details: Smart Navigation

- Consistent Nav Bar + Link to home on every page
- Clear link label
- Follow standard patterns
- Use breadcrumbs (Home > Section > Page)
- Avoid confusing jargon or menu names [1]
- 3-Click Rule
  - Not strict → Focus on user success
  - Better to go deeper if it helps clarity [2]
- Mega Menus
  - Big, hoverable menus → Lots of options, no scrolling
  - Use icons, categories, layout
  - Make parent link clickable for accessibility
- Examples: Amazon, CogniFi

## Design Details: Promotion

- Submit your site to search engines
- Add keywords in <meta> tags
- Use email carefully → Avoid spam
- Make site worth bookmarking
- Add a favicon for recognition
- Link out wisely → Give users a way back

# Integrating Collaboration: Web 2.0 Technologies

- What
  - Static pages → Interactive participation
  - Emphasizes user content and collaboration
- Key Technologies
  - **Blogs:** Article with reader comments
  - **Wikis:** Collaborative editing
  - **Social Media Links:** Facebook, Twitter, LinkedIn, YouTube
  - **Tagging/Bookmarking:** User-assigned keywords for content delivery
- Applications
  - **External:** Engage customers and partners
  - **Internal:** Boost employee collaboration

# Strategic Goals for Web 2.0 Integration

- External Goals
  - Consistent branding across platforms
  - Collect feedback and monitor opinions
  - Build online user community
  - Improve customer support and engagement
- Internal Goals
  - Foster trust and relationships
  - Share knowledge, boost innovation
  - Tag resources for easy access
  - Support company culture
- **Decision Tip:** Use tools only if aligned with user needs (found during analysis/prototyping)

# Analyst Considerations for Web 2.0

- **Align Objectives:** Balance company goals vs. user expectations
- **Be Customer's Voice:** Advocate for user needs in external designs
- **Design with Conventions**
  - Place icons where users expect (header/footer)
  - Use standard, familiar symbols
  - Follow layout norms for usability
- **Revise and Update**
  - Web tools and trends evolve fast
  - Plan for regular updates
- **Integrate with Branding**
  - Match tools with company style and voice
  - Keep look and tone consistent

# Social Media Icon Conventions

| Name of Icon Set           | Designer or Supplier | Facebook                                                                          | LinkedIn                                                                           | RSS Feed                                                                            | Twitter                                                                             |
|----------------------------|----------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Social Networks Pro Icons  | Artbees              |  |  |  |  |
| Circular Social Media      | BlogPerfume.com      |  |  |  |  |
| Free Social Media Icon Set | Elegant Themes       |  |  |  |  |
| Amazing 3-D Social Icons   | pinkmoustache.net    |  |  |  |  |
| Vintage Icons for Bloggers | Nikola Lazarevic     |  |  |  |  |

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Courtesy: Facebook, LinkedIn, and Twitter

# Introduction to App Design

- What
  - **Evolved Term:** Program → Application → App
  - Popularized by Apple; now standard
- **Design Process**
  - Begins with brainstorming
  - **UI Design Tools:** Figma, Adobe XD, Sketch, Framer, Marvel App, Proto.io, InVision, etc.
  - Developer-driven motivation; informal start
- **Focus:** Small screens (phones/tablets); principles apply across platforms

# Getting Started: Planning and Originality

- **Developer Account**
  - **Apple:** Strict approval (“walled garden”)
  - **Android:** Open platform
- **Development Process**
  - Prototyping preferred → Agile
  - Release early, iterate fast
  - Visibility boost via frequent updates
- **Be Original**
  - Don't copy
  - Avoid trademarked names

# Monetization and Branding Strategy

- **Pricing Strategy**
  - **Low-Cost:** \$0.99-\$1.99 → High volume
  - **Premium:** \$19.95+ → Niche/Pro users
  - **Freemium:** Free + IAPs
  - **Free:** Portfolio/promotional
  - **Ad-Supported:** Requires UI space
  - **Promos:** Temporary visibility boosts
- **Follow Platform Rules:** Apple's Human Interface Guidelines
- **Design the Icon**
  - Must be memorable, scalable
  - Preview tools like App Cooker
- **Choose a Name:** Max ~12 characters → Avoid truncation

# Designing for Devices and Prototyping

- **Device Screen Sizes and Icon Scaling**
  - **Challenge:** Many screen sizes even within OS
  - **Use Vector Graphics**
    - Design large
    - Scale down without quality loss
    - Great for print too
- **Design Output Screens**
  - Use drag-and-drop UI widgets
  - Follow screen design principles
- **Handle Orientation**
  - Support both Portrait and Landscape
  - Offer enhanced features in landscape
- **Share Prototypes:** Tools like AppTester for previews

# Building Functionality and Going to Market

- **Design App Logic:** Use process specs/decision trees
- **Design Navigation:** Plan screen flow/storyboards
- **Gestures**
  - Swipes, Pinches, Taps, Long presses
  - Provide feedback, consider opt-out
- **IP Protection**
  - Trademark icon/logo
  - Copyright code/design
  - Write a custom EULA
- **App Store Page**
  - Clear description
  - What's new
  - Screenshots to show best features

# References I

- [1] R. Budiu, “Information scent: How users decide where to go next,” [Online]. Available: <https://www.nngroup.com/articles/information-scent/> (cit. on p. 42).
- [2] J. Nielsen and A. Li, “Mega menus work well for site navigation,” [Online]. Available: <https://www.nngroup.com/articles/mega-menus-work-well/> (cit. on p. 42).